Grade 7 Science: Atoms & Elements - Companion Materials

PART 1: GLOSSARY & KEY TERMS

Key Terms (Alphabetical Order)

Alkali Metals (碱金属 - Jiān Jīnshǔ): Group 1 elements that are highly reactive and typically form basic (alkaline) solutions.

Alloy (合金 - Héjīn): A mixture of two or more elements, with at least one being a metal, creating a substance with improved properties.

Atom (原子 - Yuánzǐ): The basic unit of matter, consisting of a nucleus of protons and neutrons, with electrons in orbit around the nucleus.

Atomic Structure (原子结构 - Yuánzǐ Jiégòu): The arrangement of subatomic particles (protons, neutrons, and electrons) within an atom.

Chemical Symbol (化学符号 - Huàxué Fúhào): A shorthand notation used to represent an element in chemical formulas.

Compound (化合物 - Huàhéwù): A substance formed when two or more different elements chemically combine in fixed ratios.

Conductive (导电/热的 - Dàodiàn/re de): Capable of conducting electricity or heat.

Corrosion (腐蚀 - Fǔshí): The gradual degradation of materials due to chemical reactions with the environment.

Ductile (韧性 - Rènxìng): Capable of being stretched into a thin wire without breaking.

Electron (电子 - Diànzǐ): A negatively charged subatomic particle that orbits the nucleus of an atom.

Elemental Symbol (元素符号 - Yuánsù Fúhào): A symbol representing a specific chemical element.

Element (元素 - Yuánsù): A substance made up of only one type of atom, characterized by its unique number of protons.

Halogen (卤素 - Lǔsù): Group 17 elements known for their high reactivity, especially with metals.

Malleable (可锻性 - Kěduàn xìng): Capable of being shaped or bent without breaking.

Mass Number (质量数 - Zhìliàng Shù): The total number of protons and neutrons in the nucleus of an atom.

Metalloid (金属半元素 - Jīnshǔ Lèi Yuánsù): An element with properties intermediate between metals and non-metals.

Model (模型 - Móxíng): A simplified representation or description used to explain complex phenomena or structures.

Molecule (分子 - Fēnzǐ): A group of atoms bonded together, representing the smallest unit of a chemical compound.

Neutron (中子 - Zhōngzǐ): A neutral subatomic particle found in the nucleus of an atom.

Noble Gases (惰性气体 - Duòxìng Qìtǐ): Group 18 elements that are generally inert and have full electron shells.

Nucleus (原子核 - Yuánzǐ Hé): The central part of an atom, containing protons and neutrons.

Particle Diagram (粒子图表 - Lìzǐ Túbiǎo): A visual representation illustrating the arrangement and movement of particles in a substance or reaction.

Periodic Group (族 - Zú): A vertical column in the periodic table containing elements with similar chemical properties.

Periodic Period (周期 - Zhōuqī): A horizontal row in the periodic table, indicating the number of electron shells an atom has.

Periodic Table (元素周期表 - Yuán Sù Zhōuqī Biǎo): A tabular arrangement of elements based on their atomic number, electron configuration, and recurring chemical

properties.

Proton (质子 - Zhìzǐ): A positively charged subatomic particle found in the nucleus of an atom.

Proton Number (质子数 - Zhìzǐ Shù): The number of protons in the nucleus of an atom, also known as the atomic number.

Reactant (反应物 - Fǎnyìng Wù): A substance participating in a chemical reaction.

Product (产物 - Chǎnwù): A substance formed as a result of a chemical reaction.

Sulphur/Sulfur (硫磺/硫 - Liúhuáng/Liú): A non-metallic element known for its yellow color and use in various industrial processes.

Transition Metals (过渡金属 - Guòdù Jīnshǔ): Elements in groups 3-12, characterized by the filling of inner d orbitals during electron configuration.

Greek Prefixes in Chemical Nomenclature

Mono (单一 - Dānyi): Prefix indicating one in chemical names.

Di (二 - Èr): Prefix indicating two in chemical names (e.g., Carbon Dioxide).

Tri (三 - Sān): Prefix indicating three in chemical names.

Tetra (四 - Sì): Prefix indicating four in chemical names.

Penta (五 - Wǔ): Prefix indicating five in chemical names.

Hexa (六 - Liù): Prefix indicating six in chemical names.

Hepta (七 - Qī): Prefix indicating seven in chemical names.

PART 2: SUMMARY SHEETS

Compounds and Mixtures

Elements are simple substances that cannot be split up in chemical reactions. Atoms are the smallest particles of an element that can exist. Atoms in an element are all the

same. Each element has its own chemical symbol. For example, the chemical symbol for oxygen is O. The symbols and names for all the known elements are shown in the periodic table.

Some elements have their atoms joined to each other in small groups called **molecules**. Oxygen is an example.

Elements can join together to make **compounds**. A compound contains two or more elements joined together by bonds. The name of the compound tells you the elements that are in it. Compounds made from two elements always have a name that ends in '-ide'.

Many compounds exist as atoms attached to each other in small groups – molecules.

A compound always contains the same elements in the same ratio. The **chemical formula** of a compound tells you the ratio of atoms of each element that are bonded together. For compounds that are molecules it tells you the numbers of atoms of each element in a compound. Each element in the chemical formula is shown by its chemical symbol. For example: H_2O

The properties of a compound are different from the elements that make it up. For example, hydrogen is an explosive gas and oxygen will relight a glowing splint, but water is a liquid that will put fires out.

Chemical Reactions

Compounds can be made to react by mixing them with other chemicals, or by using heat or electricity. You can tell that a **chemical reaction** has occurred if there is a colour change or when a gas is given off.

Most chemical reactions also involve an energy change. This is usually in the form of heat, for example in burning (**combustion**).

In a chemical reaction a new substance is always formed. Most chemical reactions are not easily reversed (they are **irreversible**).

Some chemical reactions take place just by mixing substances together (e.g. when a solid (a **precipitate**) forms by mixing two liquids in a **precipitation reaction**). Other chemical reactions need energy to start them off (e.g. when some compounds are broken up, **decomposed**, by heat).

Word equations show what happens in a chemical reaction. The chemicals that you start with are called the **reactants**. The chemicals at the end are called the **products**. For example:

magnesium + oxygen → magnesium oxide

Physical Changes

In a physical change no new substance is formed and the changes are usually easily reversed. **Melting, evaporating, condensing and freezing** are all examples of physical changes.

Mixtures

Elements and compounds can also be mixed together. A **mixture** is easier to separate than the elements in a compound. Soil, river water and sea water are examples of mixtures that occur naturally. A pure substance contains a single substance, element or compound, nothing else.

Elements and compounds melt and boil at a fixed temperature. Mixtures do not have definite melting points and boiling points.

Alloys

Alloys are mixtures of metals with one or more other elements. Alloys have different properties from the pure metal. Pure gold is too soft for making jewellery. An alloy of gold mixed with other metals, like copper or silver, is used because it is stronger. The original method used to measure the purity of gold alloys was the carat system. One carat is 1 part in 24. So pure gold is 24 carats and 12 carat gold contains 50% gold.

PART 3: STUDENT HANDOUT & EXERCISES

Practice Questions & Exercises

The following exercises are designed to reinforce the key concepts covered in the Grade 7 Science curriculum on Atoms & Elements.

Exercise Set 1: Identifying Elements, Compounds, and Mixtures

1. Classify the following as element, compound, or mixture:
    - Pure water
    - Sea water
    - Oxygen gas
    - Iron oxide
    - Air
    - Sodium chloride
2. For each compound listed, identify the elements it contains:
    - H_2O
    - CO_2
    - NaCl
    - CuO
    - MgO
    - Fe_2O_3

Exercise Set 2: Atomic Structure

1. Complete the following table for these atoms:

Atom	Protons	Neutrons	Electrons
H			
C			
O			
Na			
Cl			

1. Calculate the mass number for each atom in the table above.

Exercise Set 3: Chemical Reactions

1. Write word equations for the following reactions:
    - Hydrogen burning in oxygen
    - Magnesium reacting with oxygen
    - Copper oxide reacting with carbon
    - Calcium carbonate decomposing with heat
2. Identify the reactants and products in each equation from question 5.

Exercise Set 4: Periodic Table

1. Using the periodic table, identify:
    - Three alkali metals
    - Three halogens
    - Three noble gases
    - Two transition metals
2. For each element identified above, state its atomic number and mass number.

Exercise Set 5: Properties of Elements

1. Classify the following elements as metals or non-metals:
    - Iron
    - Sulphur
    - Copper
    - Oxygen
    - Sodium
    - Carbon
2. For each metal identified, list three physical properties.

APPENDIX: REFERENCE TABLES

Common Elements and Their Symbols

Element	Symbol	Atomic Number	Atomic Mass
Hydrogen	H	1	1
Carbon	C	6	12
Nitrogen	N	7	14
Oxygen	O	8	16
Sodium	Na	11	23
Magnesium	Mg	12	24
Aluminium	Al	13	27
Sulphur	S	16	32
Chlorine	Cl	17	35.5
Iron	Fe	26	56
Copper	Cu	29	64
Zinc	Zn	30	65

Common Compounds and Their Formulas

Compound Name	Chemical Formula
Water	H_2O
Carbon Dioxide	CO_2
Sodium Chloride	$NaCl$
Copper Oxide	CuO
Magnesium Oxide	MgO
Iron Oxide	Fe_2O_3
Calcium Carbonate	$CaCO_3$
Sulphuric Acid	H_2SO_4
Sodium Sulphate	Na_2SO_4
Copper Sulphate	$CuSO_4$

End of Companion Materials